

Fiscal Backing and Heterogeneous Household Portfolios^{*}

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Abstract

When do fiscal deficits become inflationary? I answer this question in a heterogeneous-agent New Keynesian model in which households hold liquid nominal government liabilities and illiquid real capital. The answer depends on fiscal backing. Deficits that are not backed by higher expected future primary surpluses generate much larger inflation and output responses than backed deficits, because weak backing induces households to reduce their demand for nominal government liabilities and rebalance toward real capital. This portfolio-reallocation channel makes unfunded government purchases and unfunded transfers produce nearly identical responses, while funded deficits depend more on the fiscal instrument. Easier adjustment of illiquid assets enlarges the gap between funded and unfunded deficits. The results show that portfolio choice makes fiscal backing quantitatively central for inflation.

JEL codes: E62, E31, E52, D31, G11

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Introduction

Large fiscal expansions and elevated inflation after the COVID-19 pandemic have renewed interest in a basic question: under what circumstances do fiscal deficits raise inflation? A fiscal deficit does not, by itself, imply inflation. When higher government debt is expected to be matched by higher future primary surpluses, the fiscal expansion mainly changes the timing of fiscal adjustment rather than the valuation of government debt. Fiscal explanations of inflation require that deficits have real effects that are not neutralized by expected future surpluses.

One channel comes from the Fiscal Theory of the Price Level (FTPL). The real value of outstanding nominal government debt must be consistent with expected future fiscal backing. When higher deficits are not expected to be matched by higher future primary surpluses, the price level rises to reduce the real value of nominal government debt until household asset demand and fiscal backing are again consistent with equilibrium ([Cochrane, 2001](#); [Leeper, 1991](#); [Woodford, 1995](#)). This logic has been used to interpret both earlier episodes of fiscally driven inflation and the post-COVID inflation surge ([Balke and Zarazaga, 2024](#); [Barro, 2022](#); [Bianchi et al., 2023](#); [Bianchi and Ilut, 2017](#); [Hall and Sargent, 2011](#)).

An alternative channel comes from heterogeneous-household models with nominal rigidities. Fiscal deficits become inflationary when they shift resources toward households with high marginal propensities to consume (MPCs), whose consumption responds strongly to additional income. The resulting rise in aggregate demand generates quantitatively important inflation responses even without making fiscal backing central to the analysis ([Angeletos et al., 2024](#)). This literature suggests that heterogeneity alone can account for substantial inflationary effects of fiscal deficits. That conclusion, however, is derived in environments in which households save in a single liquid nominal government bond and do not choose whether to reallocate wealth into illiquid real capital.

I ask whether redistribution alone remains sufficient to generate large inflation responses once households can also shift wealth between liquid nominal and illiquid real assets. In

environments with only a liquid nominal government asset, households can respond to weak fiscal backing only by changing consumption and nominal saving. In this paper, they can also rebalance away from nominal government liabilities and toward real capital. This additional margin is central because it changes how fiscal backing is transmitted to inflation and real activity.

Building on the framework in [Bayer et al. \(2024\)](#), I develop a heterogeneous-agent New Keynesian (HANK) model with long-term nominal government debt, sticky prices and wages, and endogenous household portfolio choice. Households can save or borrow liquid nominal assets, but can also hold illiquid real capital that can be adjusted only intermittently. Fiscal policy operates through government purchases and transfers, including both uniform lump-sum transfers and transfers targeted to different household groups. I compare funded deficits, which are backed by higher expected future primary surpluses, with unfunded deficits, for which future primary surpluses do not provide sufficient backing.

The main findings are fivefold. First, unfunded deficits generate much larger aggregate responses than funded deficits. When future primary surpluses do not provide sufficient fiscal backing, households reduce their demand for nominal government liabilities and rebalance their portfolios toward illiquid real capital. This portfolio reallocation amplifies the decline in the real value of nominal government debt and generates larger responses of investment, output, and inflation. Second, when deficits are unfunded, government purchases and transfers generate very similar aggregate responses. This result suggests that fiscal backing itself, rather than the specific fiscal instrument, is the key determinant of the aggregate inflationary effects of deficits. Third, unfunded fiscal shocks lower consumption inequality by raising the consumption of households in the bottom and middle of the distribution and lowering the consumption of households at the top. The decline in the real value of nominal government debt reduces the wealth of households with large nominal asset positions, while higher output and transfers support households with lower wealth. Fourth, when deficits are funded, the type of fiscal expansion matters. Because funded deficits are backed by higher expected fu-

ture primary surpluses, they do not trigger a large portfolio reallocation away from nominal government bonds to the same extent as unfunded deficits do. Redistribution through the fiscal instrument becomes more important: funded transfers provide immediate support to households with high marginal propensities to consume, while funded government purchases raise the consumption of households with low marginal propensities to consume.

Fifth, lessening the frictions that restrict households' access to the illiquid capital market enlarges the quantitative gap between the responses to funded and unfunded deficits: unfunded deficits generate larger responses, while funded deficits generate smaller ones. A decline in asset-holding frictions has two effects. First, households can self-insure idiosyncratic risk more effectively and smooth consumption more over time. This dampens responses to funded deficits, because households have less need to front-load consumption out of additional disposable resources. Second, households can rebalance wealth away from nominal government bonds more easily when fiscal backing weakens. For unfunded deficits, the decline in the real value of nominal government bonds creates a stronger incentive to shift wealth toward illiquid real assets, raising investment and thereby increasing output and inflation. This result suggests that household heterogeneity alone does not replace the FTPL explanation of inflation. Once households can reallocate wealth across liquid nominal bonds and illiquid real assets, fiscal backing remains quantitatively central for inflation.

This paper contributes to two primary strands of literature. First, it speaks to the growing literature on fiscal deficits, inflation, and the FTPL in heterogeneous-agent environments. The FTPL literature shows that the real value of nominal government debt depends on expected fiscal backing ([Bianchi et al., 2023](#); [Bianchi and Ilut, 2017](#); [Cochrane, 2001](#); [Hall and Sargent, 2011](#); [Leeper, 1991](#); [Woodford, 1995](#)). Closest in spirit, [Leeper et al. \(2017\)](#) study fiscal financing and monetary-fiscal regimes in an estimated DSGE model with savers and non-savers, focusing on government spending multipliers across regimes. Recent work brings fiscal backing into richer heterogeneous-agent environments: [Kaplan et al. \(2026\)](#) develop a heterogeneous-agent fiscal theory of the price level, but in a flexible-price economy

in which households save in a single liquid nominal government asset. Other recent work emphasizes a redistribution channel: deficits can raise inflation by shifting resources toward households with high marginal propensities to consume under nominal rigidities ([Angeletos et al., 2024](#); [Kwicklis, 2025](#)). These papers show that household heterogeneity matters for fiscal inflation, but they do not study how fiscal backing is transmitted when households can choose between liquid nominal government liabilities and illiquid real capital.

In the current paper, I add that missing portfolio-reallocation margin. Once households can choose between nominal government liabilities and real capital, redistribution alone does not fully account for the inflationary effects of fiscal deficits. When deficits are unfunded, weak fiscal backing induces households to reduce their demand for nominal government liabilities and shift wealth toward real assets. This channel makes unfunded deficits much more inflationary than funded deficits and makes unfunded government purchases and unfunded transfers generate similar aggregate responses. The paper therefore raises doubt about the view that the inflationary effects of deficits are primarily determined by MPC heterogeneity or fiscal targeting. Whether the deficit is fiscally backed becomes the central determinant of the aggregate response to fiscal shocks.

Second, I contribute to the literature on household portfolios and policy transmission. Quantitative work with liquid and illiquid assets shows that household portfolio structure matters for aggregate dynamics and policy transmission ([Bayer et al., 2023, 2024](#); [Luetticke, 2021](#)). I use this liquid-illiquid portfolio structure to study when fiscal deficits become inflationary. Lowering frictions in the illiquid capital market enlarges the difference between funded and unfunded deficits: funded deficits generate smaller responses because households can smooth consumption more effectively, while unfunded deficits generate larger responses because households can rebalance away from nominal government bonds more easily when fiscal backing weakens. This result shows that household heterogeneity and portfolio choice do not replace the FTPL explanation of inflation. Instead, portfolio choice makes fiscal backing more important by determining how strongly households can move away from nominal

government liabilities when backing weakens.

The remainder of the paper is organized as follows. Section 1 presents the model. Section 2 discusses the calibration. Section 3 reports the quantitative results. Section 4 concludes.

1 Model

I build a heterogeneous-agent New Keynesian model with incomplete markets, long-term nominal government debt, and endogenous household portfolio choice. Households save in two assets: a liquid nominal asset and illiquid real capital. The government runs deficits through government purchases and transfers and distinguishes between funded deficits, for which expected future primary surpluses provide sufficient backing, and unfunded deficits, for which expected future primary surpluses do not offset current deficits. I consider transfers may be distributed uniformly across households or are targeted to specific household groups. Firms operate under monopolistic competition with sticky prices and wages. Banks intermediate household liquid assets into long-term nominal government debt.

The household side builds on [Bayer et al. \(2024\)](#), the banking and long-term debt side follows [Bayer et al. \(2023\)](#), and the funded-versus-unfunded fiscal structure follows [Balke and Zarazaga \(2024\)](#). Appendix A provides additional details on preferences, taxation, firm-side optimization, and wage setting.

1.1 Government and Fiscal Backing

The government issues long-term nominal debt, levies taxes on labor and profit income, and runs deficits through government purchases and transfers. I distinguish between funded government purchases and transfers, for which expected future primary surpluses provide sufficient backing, and unfunded government purchases and transfers, for which expected future primary surpluses do not provide sufficient backing. Monetary policy follows a passive

Taylor rule.

1.1.1 Fiscal Authority

Let $Bgov_{t+1}$ denote total government debt at the end of period t . I decompose total government debt into funded debt, $BFgov_{t+1}$, and unfunded debt, $BUGov_{t+1}$:

$$Bgov_{t+1} = BFgov_{t+1} + BUGov_{t+1}. \quad (1)$$

Total government debt evolves according to

$$Bgov_{t+1} = G_t + Transfer_t - Tax_t + \frac{R_t^b}{\pi_t} Bgov_t, \quad (2)$$

where G_t is total government consumption, $Transfer_t$ is total government transfers, Tax_t is tax revenue, R_t^b is the gross nominal interest rate on government debt, and π_t is gross inflation.

Funded debt evolves according to

$$BFgov_{t+1} = GF_t + TF_t - \chi_1 Tax_t + \frac{R_t^b}{\pi_t} BFgov_t, \quad (3)$$

where GF_t and TF_t denote the funded components of government purchases and transfers, and $\chi_1 \in [0, 1]$ is the share of tax revenue allocated to the funded debt component. I set $\chi_1 = BF\bar{gov}/B\bar{gov}$ in steady state. The unfunded component is then

$$BUGov_{t+1} = Bgov_{t+1} - BFgov_{t+1}. \quad (4)$$

Government purchases and transfers are split into funded and unfunded components:

$$G_t = GF_t + GU_t, \quad (5)$$

$$Transfer_t = TF_t + TU_t. \quad (6)$$

Out of steady state, the four components evolve according to

$$\frac{GU_t}{\bar{GU}} = \left(\frac{GU_{t-1}}{\bar{GU}} \right)^{\rho_{GU}} \left(\frac{Y_t}{Y_{t-1}} \right)^{(1-\rho_{GU})\gamma_{YGU}} \left(\frac{BUgov_t}{\bar{BUgov}} \right)^{(1-\rho_{GU})\gamma_{BGU}} \epsilon_t^{GU}, \quad (7)$$

$$\frac{GF_t}{\bar{GF}} = \left(\frac{GF_{t-1}}{\bar{GF}} \right)^{\rho_{GF}} \left(\frac{Y_t}{Y_{t-1}} \right)^{(1-\rho_{GF})\gamma_{YGF}} \left(\frac{BFgov_t}{\bar{BFgov}} \right)^{(1-\rho_{GF})\gamma_{BGF}} \epsilon_t^{GF}, \quad (8)$$

$$\frac{TU_t}{\bar{TU}} = \left(\frac{TU_{t-1}}{\bar{TU}} \right)^{\rho_{TU}} \left(\frac{Y_t}{Y_{t-1}} \right)^{(1-\rho_{TU})\gamma_{YTU}} \left(\frac{BUgov_t}{\bar{BUgov}} \right)^{(1-\rho_{TU})\gamma_{BTU}} \epsilon_t^{TU}, \quad (9)$$

$$\frac{TF_t}{\bar{TF}} = \left(\frac{TF_{t-1}}{\bar{TF}} \right)^{\rho_{TF}} \left(\frac{Y_t}{Y_{t-1}} \right)^{(1-\rho_{TF})\gamma_{YTF}} \left(\frac{BFgov_t}{\bar{BFgov}} \right)^{(1-\rho_{TF})\gamma_{BTF}} \epsilon_t^{TF}. \quad (10)$$

The parameters γ_{BG_-} and γ_{BT_-} measure how strongly government purchases and transfers respond to debt. A higher value implies that higher debt is followed by a larger increase in the corresponding spending component. This makes it harder for future primary surpluses to rise enough to provide sufficient backing for the debt. Lower values therefore correspond to sufficient fiscal backing, whereas higher values correspond to insufficient fiscal backing. In the calibration, the funded coefficients, γ_{BGF} and γ_{BTF} , are set low enough that fiscal policy remains passive, whereas the unfunded coefficients, γ_{BGU} and γ_{BTU} , are set high enough that fiscal policy is active.

Taxation. The tax system follows [Bayer et al. \(2024\)](#). The average tax rate τ_t evolves according to

$$\frac{\tau_t}{\bar{\tau}} = \left(\frac{\tau_{t-1}}{\bar{\tau}} \right)^{\rho_{\tau}} \left(\frac{BFgov_t}{\bar{BFgov}} \right)^{(1-\rho_{\tau})\gamma_B^{\tau}} \left(\frac{Y_t}{Y_{t-1}} \right)^{(1-\rho_{\tau})\gamma_Y^{\tau}}, \quad (11)$$

and tax progressivity follows

$$\frac{\tau_t^P}{\bar{\tau}^P} = \left(\frac{\tau_{t-1}^P}{\bar{\tau}^P} \right)^{\rho_P} \epsilon_t^P. \quad (12)$$

Aggregate tax revenue is denoted by Tax_t . Tax revenue is raised from labor income and profits and feeds back on funded debt. Appendix A.1 describes the household-level tax structure in detail.

1.1.2 Monetary Authority

The monetary authority sets the nominal interest rate according to a Taylor-type rule,

$$\frac{R_{t+1}^b}{\bar{R}^b} = \left(\frac{R_t^b}{\bar{R}^b} \right)^{\rho_R} \left(\frac{\pi_t}{\bar{\pi}} \right)^{(1-\rho_R)\theta_\pi} \left(\frac{Y_t}{Y_{t-1}} \right)^{(1-\rho_R)\theta_Y} \epsilon_t^R. \quad (13)$$

I focus on a passive monetary regime in the sense of [Leeper \(1991\)](#), $\theta_\pi < 1$.

1.2 Households and Portfolio Choice

A continuum of households faces idiosyncratic labor-income risk and chooses how much to save in a liquid nominal asset and in illiquid real capital. Workers earn labor income and receive union profits. Entrepreneurs do not work but do receive firm profits.

1.2.1 Income, Assets, and Budget Constraint

The household side follows [Bayer et al. \(2024\)](#). Workers face idiosyncratic productivity risk. Entrepreneurs do not supply labor and receive non-union firm profits. Let b_{it} denote household i 's liquid asset holdings and k_{it} denote illiquid real capital. Households can borrow through the liquid asset up to the limit $\underline{B}$, while illiquid capital must remain nonnegative.

Household i 's period- t budget constraint is

$$c_{it} + b_{it+1} + q_t k_{it+1} = \frac{R_{it}}{\pi_t} b_{it} + (q_t + r_t) k_{it} + y_{it} + \mathbb{I}_{h_{it} \neq 0} (1 - \tau_t) \Pi_t^U + \mathbb{I}_{h_{it} = 0} (1 - \tau_t^L) (\Pi_t^E)^{1-\bar{\tau}^P} + Transfers_{it}, \quad (14)$$

subject to

$$k_{it+1} \geq 0, \quad b_{it+1} \geq \underline{B}. \quad (15)$$

Here, q_t is the price of capital, r_t is the dividend on capital, y_{it} is after-tax labor income, Π_t^U denotes union profits received by workers, Π_t^E denotes entrepreneurial profits, and $Transfers_{it}$ denotes government transfers.

Households choose b_{it+1} rather than direct holdings of long-term government debt. The return on the liquid asset is

$$R_{it} = \begin{cases} A_t \frac{R_t^b B_t + \pi_t [(1 - \iota^\Pi) q_t^\Pi + \omega^\Pi \Pi_t^F]}{B_t + q_{t-1}^\Pi} & \text{if } b_{it} \geq 0, \\ A_t \frac{R_t^b B_t + \pi_t [(1 - \iota^\Pi) q_t^\Pi + \omega^\Pi \Pi_t^F]}{B_t + q_{t-1}^\Pi} + \bar{R} & \text{if } b_{it} < 0. \end{cases} \quad (16)$$

Here, A_t is an aggregate liquid-return wedge. The liquid asset return equals the average return on the aggregate liquid portfolio, composed of government debt and tradable claims on firm profits, plus a borrowing wedge for households with negative liquid asset holdings.

1.2.2 Portfolio Adjustment Friction

Only a fraction λ of households can adjust illiquid capital in a given period. Households that receive the adjustment opportunity choose both liquid assets and illiquid capital. Households that do not receive the adjustment opportunity keep illiquid capital fixed and choose only liquid asset holdings. The parameter λ governs how easily households reallocate wealth between liquid nominal assets and illiquid real capital.

1.2.3 Household Problem

Let $\Theta_t(b, k, h)$ denote the joint distribution of liquid asset holdings, capital, and idiosyncratic productivity. Define $V_t^a(b, k, h)$ as the value function for households that can adjust both

assets, and $V_t^n(b, k, h)$ as the value function for households that cannot adjust capital:

$$V_t^a(b, k, h) = \max_{b'_a, k'} u[x(b, b'_a, k, k', h)] + \beta \mathbb{E}_t \mathbb{W}_{t+1}(b'_a, k', h'), \quad (17)$$

$$V_t^n(b, k, h) = \max_{b'_n} u[x(b, b'_n, k, k, h)] + \beta \mathbb{E}_t \mathbb{W}_{t+1}(b'_n, k, h'), \quad (18)$$

$$\mathbb{W}_{t+1}(b', k', h') = \lambda V_{t+1}^a(b', k', h') + (1 - \lambda) V_{t+1}^n(b', k', h'). \quad (19)$$

The cross-sectional distribution evolves according to household policy rules and the transition matrix for idiosyncratic productivity:

$$\begin{aligned} \Theta_{t+1}(b', k', h') &= \lambda \int_{b'=b_{a,t}^*(b,k,h), k'=k_t^*(b,k,h)} \Phi(h, h') d\Theta_t(b, k, h) \\ &\quad + (1 - \lambda) \int_{b'=b_{n,t}^*(b,k,h), k'=k} \Phi(h, h') d\Theta_t(b, k, h). \end{aligned} \quad (20)$$

Appendix [A.1](#) provides additional details on preferences, labor supply, tax progressivity, and the derivation of household and aggregate income.

1.3 Firms, Banks, and Long-Term Government Debt

1.3.1 Firms

The production side follows a standard New Keynesian structure with monopolistic competition, sticky prices, sticky wages, and investment adjustment costs. Intermediate firms produce using labor and capital,

$$Y_t = Z_t N_t^{1-\alpha} (u_t K_t)^\alpha, \quad (21)$$

where Z_t is total factor productivity, N_t is labor input, K_t is the capital stock, and u_t is capital utilization.

Price setting and wage setting imply a standard price Phillips curve and wage Phillips

curve. Capital accumulation is subject to convex adjustment costs. Appendix [A.4](#) provides the firm-side details.

1.3.2 Banks and Long-Term Nominal Debt

Banks intermediate aggregate household liquid asset holdings into long-term nominal government debt, following [Bayer et al. \(2023\)](#). Government bonds have geometric maturity. Each bond pays one unit of nominal coupon per period, and a fraction δ_B matures each period. Let q_t^B denote the nominal bond price. Perfect competition in banking implies

$$R_{t+1}^b q_t^B = \mathbb{E}_t [(1 - \delta_B) q_{t+1}^B + 1]. \quad (22)$$

Real bank profits are

$$\Pi_t^B = \frac{Bgov_t}{\pi_t} \left[(1 - \delta_B) \frac{q_t^B}{q_{t-1}^B} + \frac{1}{q_{t-1}^B} - R_t^b \right]. \quad (23)$$

Banks map household liquid asset demand into the pricing of long-term nominal government debt.

1.4 Market Clearing and Equilibrium

Liquid asset market. The liquid asset market clears when

$$B_{t+1} = \mathbb{E}_t [\lambda b_{a,t}^* + (1 - \lambda) b_{n,t}^*]. \quad (24)$$

Aggregate liquid asset supply consists of long-term government debt and tradable profit claims. With the timing convention in the model, household liquid asset holdings carried into period t equal the government debt carried into period t plus the market value of tradable

profit claims purchased in period $t - 1$:

$$B_t = Bgov_t + q_{t-1}^\Pi, \quad (25)$$

where q_{t-1}^Π is the price of tradable profit claims at the end of period $t - 1$. Since the supply of tradable profit claims is normalized to one, q_{t-1}^Π is also their aggregate market value.

Tradable profit claims. The price of tradable profit claims satisfies

$$q_t^\Pi R_t^b = \mathbb{E}_t \pi_{t+1} [(1 - \iota^\Pi) q_{t+1}^\Pi + \omega^\Pi \Pi_{t+1}^F]. \quad (26)$$

Here, ι^Π is the fraction of outstanding tradable profit claims that matures each period, so $1 - \iota^\Pi$ is the fraction of claims that remains outstanding into the next period. The parameter ω^Π denotes the share of firm profits paid to tradable profit-claim holders.

Capital market. The capital market clears when aggregate household demand for illiquid capital equals the aggregate capital stock. Households that can adjust illiquid assets choose $k_t^*(b, k, h)$, while non-adjusting households keep their current illiquid capital position k_t . Thus,

$$K_{t+1} = \int [\lambda k_t^*(b, k, h) + (1 - \lambda) k_t] d\Theta_t(b, k, h), \quad (27)$$

where $\Theta_t(b, k, h)$ is the joint distribution of liquid asset holdings, illiquid capital, and idiosyncratic productivity at the beginning of period t .

Goods and labor markets. The labor market clears at the equilibrium real wage. The goods market clears through the resource constraint. Market clearing in liquid assets, capital, and goods jointly determines how changes in fiscal backing translate into inflation.

A sequential equilibrium consists of household policy functions, a law of motion for the cross-sectional distribution, sequences of prices and aggregates, and government policies

such that households optimize, firms and banks optimize, government budget constraints and policy rules hold, and all markets clear.

2 Calibration

I calibrate the model at quarterly frequency. The steady state is chosen to match the same broad long-run objects as [Bayer et al. \(2024\)](#), with particular attention to household portfolio structure, the prevalence of borrowers, long-term nominal debt, and the fiscal side. The targeted moments include earnings persistence and dispersion, worker–entrepreneur transitions, the liquid–illiquid composition of household wealth, factor shares, government size, and tax progressivity.

2.1 Steady-State Calibration

The household side is disciplined by a small set of preference, income-process, and portfolio parameters. I set

$$(\xi, \gamma, \beta, \gamma_{\text{scale}}) = (4, 2, 0.98255, 0.2),$$

which pin down precautionary saving, labor supply, the capital–output ratio, and steady-state hours. The idiosyncratic income process is governed by $(\rho_h, \sigma_h) = (0.98, 0.12)$, and worker–entrepreneur transitions are set to $\iota = 1/16$ and $\zeta = 1/4500$, so that the model matches earnings persistence, earnings dispersion, and a thin entrepreneurial tail.

The portfolio structure is disciplined by three key parameters. The probability of illiquid asset adjustment is $\lambda = 0.03$, which controls how often households can rebalance between liquid nominal assets and illiquid capital. Long-term government debt has geometric maturity with $\delta_B = 1/5$, which implies an average duration of five quarters. The relative supply of liquid assets is governed by $\psi = 0.10$, and tradable profit claims are set by $(\omega_\Pi, \iota_\Pi) = (0.15, 0.016)$. Together, these parameters pin down the liquid–illiquid composition of household portfolios and the prevalence of borrowers.

Table 1: Key calibrated parameters

Category	Symbol	Value	Role / target
Preferences	ξ	4.0	Precautionary saving
	γ	2.0	Labor supply elasticity
	β	0.98255	K/Y
Income risk	ρ_h	0.98	Earnings persistence
	σ_h	0.12	Earnings dispersion
	ι, ζ	1/16, 1/4500	Worker–entrepreneur flows
Portfolio structure	λ	0.03	Illiquid asset adjustment frequency
	δ_B	0.20	Debt maturity
	ψ	0.10	Liquid asset supply
	ω_Π, ι_Π	0.15, 0.016	Tradable profit-claim supply
Technology	α	0.318	Labor share
	δ_0	0.0215	Depreciation
	μ^Y, μ^W	1.1, 1.1	Price and wage markups
Fiscal side	τ^L, τ^P	0.645, 0.1022	Government size and progressivity
	$\bar{R}$	0.0466	Borrowing wedge

On the production side, I set the capital share to $\alpha = 0.318$, consistent with a labor share of about 0.68, and quarterly depreciation to $\delta_0 = 0.0215$. Gross price and wage markups are set to $\mu^Y = \mu^W = 1.1$. On the fiscal side, tax parameters are $(\tau^L, \tau^P) = (0.645, 0.1022)$, which imply a government consumption share near 20 percent and a realistic degree of tax progressivity. I normalize steady-state gross inflation and the gross policy rate to $\pi = R^b = 1$. The borrower wedge is $\bar{R} = \pi(1.20)^{1/4} - 1 \approx 0.0466$.

Table 1 summarizes the main calibrated parameters used in the quantitative exercises. Appendix Table 3 reports the full steady-state calibration, targets, and data sources.

2.2 Policy Rules for Funded and Unfunded Deficits

Funded and unfunded deficits are implemented through different debt-feedback coefficients in the fiscal rules. Since the rules are written for government purchases and transfers, a negative debt-feedback coefficient means that higher debt reduces purchases or transfers. This raises primary surpluses and therefore funds the initial deficit. A positive debt-feedback coefficient

Table 2: Baseline policy-rule parameters

Rule	Parameters	Values
Funded government purchases	$(\rho_{GF}, \gamma_{YGF}, \gamma_{BGF})$	$(0.90, 0.50, -0.50)$
Unfunded government purchases	$(\rho_{GU}, \gamma_{YGU}, \gamma_{BGU})$	$(0.90, 0.50, 0.50)$
Funded transfers	$(\rho_{TF}, \gamma_{YTF}, \gamma_{BTF})$	$(0.90, 0.50, -0.50)$
Unfunded transfers	$(\rho_{TU}, \gamma_{YTU}, \gamma_{BTU})$	$(0.90, 0.50, 0.50)$
Monetary policy	$(\rho_R, \theta_\pi, \theta_Y)$	$(0.70, 0.50, 0.125)$

means that higher debt is followed by higher purchases or transfers, so the initial deficit is not backed by primary surpluses.

In the baseline calibration, funded government purchases and transfers load on negative debt-feedback coefficients, $\gamma_{BGF} = \gamma_{BTF} = -0.50$, while unfunded government purchases and transfers load on positive debt-feedback coefficients, $\gamma_{BGU} = \gamma_{BTU} = 0.50$.^{*} The other fiscal-rule coefficients are held fixed across the four deficit experiments: all government-purchase and transfer rules have persistence $\rho = 0.90$ and output response $\gamma_Y = 0.50$. Thus, the main difference between funded and unfunded experiments comes from the debt-feedback coefficient. Table 2 reports the baseline fiscal-rule coefficients, the monetary policy rule, and the shock processes used in the quantitative exercises.

All exogenous shocks follow AR(1) processes. For the benchmark impulse responses, I set the innovation standard deviations to a common small value and scale the reported responses to a common shock size. Appendix Table 5 reports the full set of policy-rule and shock-process parameters.

2.3 Solution Method

I solve the model following the Reiter-style approach in Bayer et al. (2024). I first compute the stationary equilibrium by iterating on household policy functions and the invariant cross-sectional distribution. I then reduce the high-dimensional distributional state, linearize the equilibrium system around the stationary equilibrium, and solve the resulting linear

^{*}The results are robust to setting $\gamma_{BGU} = \gamma_{BTU} = 0$.

rational-expectations system to obtain impulse responses. Appendix C reports the numerical algorithm, state reduction procedure, and diagnostic checks.

3 Results

In this section, I examine the responses of aggregate variables to funded and unfunded fiscal shocks to transfers and government purchases. I also examine the distributional consequences of these shocks.

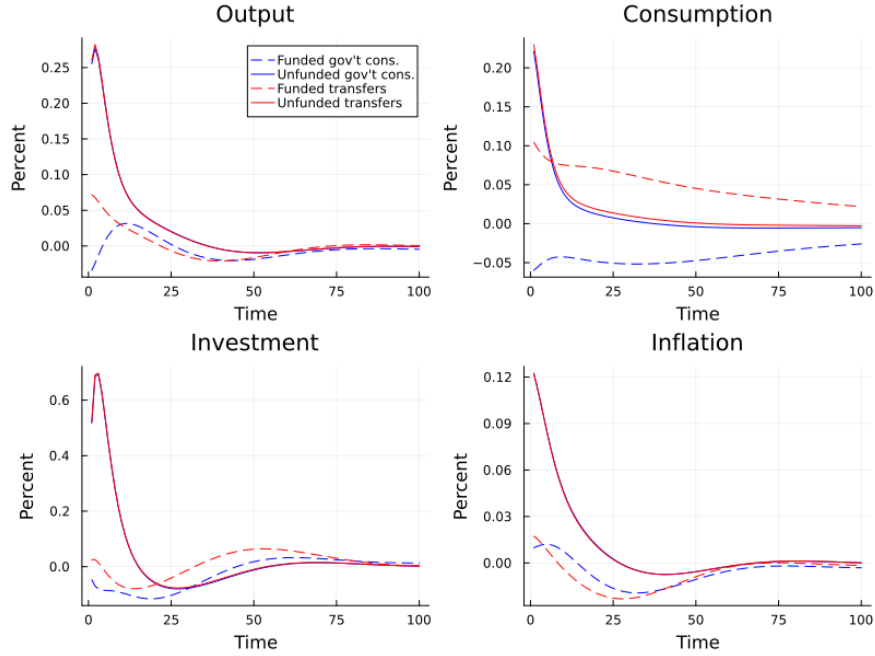
3.1 Aggregate Effects on Output, Consumption, Investment, and Inflation

This section reports impulse response functions for four fiscal deficit shocks. The shocks differ along two dimensions: whether the deficit comes from government purchases or transfers, and whether the deficit is funded or unfunded.

Figure 1 reports the impulse responses of output, consumption, investment, and inflation to fiscal shocks. Two results stand out. First, unfunded fiscal shocks generate much larger responses in all four variables than funded fiscal shocks. In the figure, the unfunded responses are several times larger than the corresponding funded responses, while the funded responses remain relatively modest. Second, among unfunded shocks, government purchases and transfers generate very similar responses. In fact, the difference in responses is not discernible in the results.

The contrast between funded and unfunded shocks comes from fiscal backing. When the deficit is funded, higher expected future primary surpluses sustain household demand for nominal government bonds. The shift away from nominal bonds is therefore limited, goods demand rises by less, and output, consumption, investment, and inflation move only modestly. When the deficit is unfunded, weak backing reduces household demand for nominal government bonds and shifts demand toward goods. As a result, real activity increases more

Figure 1: Impulse Response Functions (IRFs) to fiscal deficit shocks



Note: Impulse Response Functions (IRFs) for output, consumption, investment, and inflation, showing the effects of a 0.1-percent temporary increase in fiscal expenditures. Solid lines represent unfunded shocks, and dashed lines represent funded shocks. Red lines indicate transfers, and blue lines indicate government consumption. Transfer shocks are distributed evenly across households.

strongly and the price level must increase by more as well.

This same mechanism explains why unfunded government-purchases shocks and unfunded transfer shocks look so similar. Once fiscal backing is weak, the common portfolio adjustment away from nominal government bonds becomes the dominant force. Differences between purchases and transfers matter less for aggregate dynamics, so the two unfunded shocks produce nearly identical responses.

By contrast, when the deficit is funded, the source of the deficit matters more. A funded transfer raises household disposable income directly, so its effect depends on who receives the transfer and how strongly those households spend out of additional income. Transfers work through households' marginal propensities to consume: a larger share of the transfer is spent when it reaches households with high MPCs, while more of it is saved when it reaches

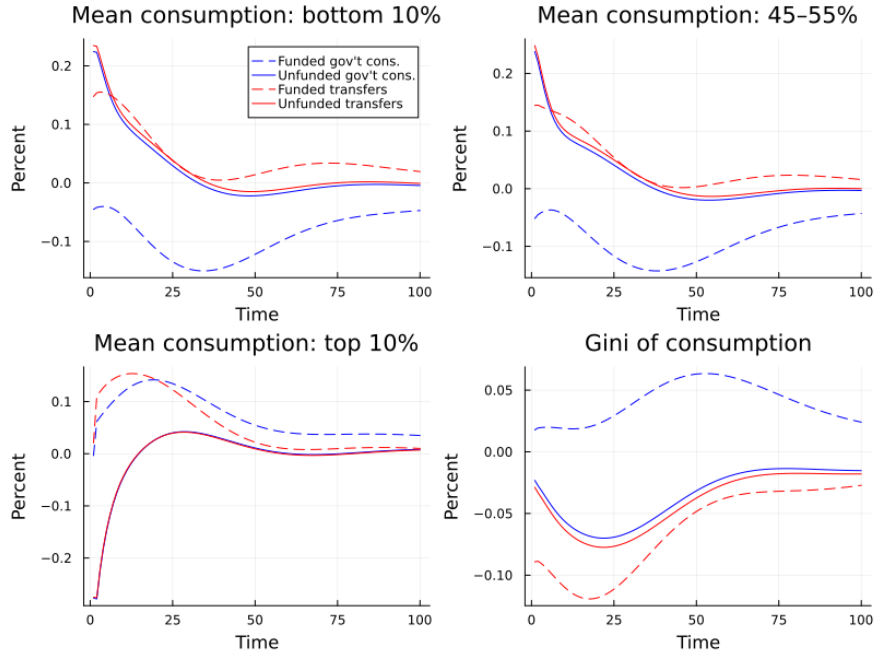
households with low MPCs. A funded government-purchases shock operates differently. It raises demand because the government directly purchases goods, rather than because households receive additional disposable income. Since the deficit is funded, however, the increase in government purchases is backed by higher expected future primary surpluses. Households do not treat the fiscal expansion as an unbacked increase in nominal wealth. Future fiscal adjustment limits the rise in private spending and can crowd out consumption and investment. At the same time, sufficient fiscal backing prevents the large portfolio reallocation away from nominal government bonds that occurs under unfunded deficits. The direct increase in government demand is partly offset by weaker private demand. In the calibration, funded government-purchases shocks are the least expansionary case.

3.2 Distributional Effects: Mean Consumption Across the Liquid-Asset Distribution and Consumption Inequality

Figure 2 reports the impulse responses of mean consumption in the bottom 10 percent, the 45th–55th percentiles, and the top 10 percent of the liquid-asset distribution, together with the consumption Gini. The liquid-asset distribution ranks households by their holdings of liquid nominal assets. Households at the bottom of this distribution hold little liquid wealth or are borrowing constrained, while households at the top hold large positive liquid nominal positions. Transfer shocks are lump-sum and distributed evenly across households.

Unfunded government-purchases shocks and unfunded transfer shocks raise mean consumption strongly in the bottom 10 percent and in the 45th–55th percentiles, while reducing mean consumption for the top 10 percent. Funded transfers also raise mean consumption for households with low liquid assets. Funded government-purchases shocks, by contrast, lower mean consumption in the bottom 10 percent and in the 45th–55th percentiles, while raising it for the top 10 percent. The reason is that funded government purchases raise demand directly through the government sector, but do not provide households with an immediate increase in disposable income. Since the deficit is backed by higher expected future primary

Figure 2: Aggregate responses to fiscal deficit shocks: mean consumption across the liquid-asset distribution and consumption inequality



Note: Impulse Response Functions (IRFs) for mean consumption in the bottom 10 percent, the 45th–55th percentiles, and the top 10 percent of the liquid-asset distribution, together with the consumption Gini, following a 0.1-percent temporary increase in fiscal expenditures. Solid lines represent unfunded shocks, and dashed lines represent funded shocks. Red lines indicate transfers, and blue lines indicate government consumption. Transfer shocks are lump-sum and distributed evenly across households.

surpluses, households also anticipate future fiscal adjustment. This limits the revaluation of nominal debt, weakens the portfolio-reallocation channel, and creates crowding out of private consumption. In the calibration, this crowding out falls more heavily on households with low and middle liquid-asset holdings, while households at the top are less exposed to the decline in disposable resources and benefit relatively more from their asset positions and profit-related income. The consumption Gini falls under both unfunded shocks and under funded transfers, but rises under funded government purchases.

When deficits are unfunded, mean consumption rises strongly in the bottom 10 percent and in the 45th–55th percentiles because aggregate demand and income expand. For house-

holds at the top of the liquid-asset distribution, the dominant force is different. Since they hold large positive nominal positions, inflation reduces the real value of those assets and lowers their purchasing power. Mean consumption therefore falls for the top 10 percent. Consumption inequality declines because the shock raises consumption for households with low and middle liquid wealth while reducing consumption for households with high liquid wealth.

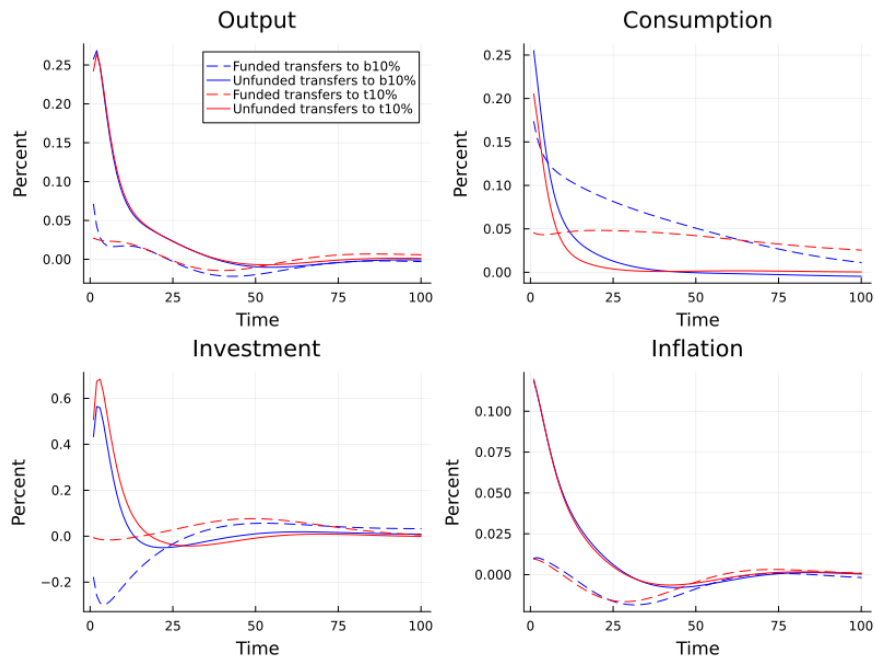
When deficits are funded, the source of the deficit matters more. Funded transfers raise disposable income directly and support households with low liquid assets more strongly. Since transfers are distributed evenly across households, the same dollar transfer represents a larger effective increase in resources for households with little liquid wealth. These households also tend to have higher MPCs, so their consumption responds more strongly. As a result, funded transfers reduce consumption inequality.

Funded government purchases operate differently. They raise government demand directly but do not provide an immediate lump-sum increase in household disposable income. Because the deficit is funded, expected future primary surpluses limit the revaluation of nominal debt and mute portfolio reallocation away from nominal government bonds. In the calibration, the implied future fiscal adjustment and the absence of direct household transfers lower mean consumption in the bottom 10 percent and in the 45th–55th percentiles, while the top 10 percent benefits relatively more from the associated movements in asset returns and aggregate income. Funded government purchases are therefore the only case in which the consumption Gini rises persistently.

3.3 Impact of Transfer Targeting and Fiscal Backing

Figure 3 reports the impulse responses of output, consumption, investment, and inflation to transfer shocks targeted to either the bottom 10 percent or the top 10 percent of the liquid-asset distribution. In each case, each recipient receives the same amount. Two results stand out. First, when transfers are funded, targeting matters a great deal for consumption and

Figure 3: Aggregate responses to targeted funded and unfunded transfer shocks



Note: Impulse Response Functions (IRFs) for output, consumption, investment, and inflation, showing the effects of a 0.1-percent temporary increase in fiscal transfers. Solid lines represent unfunded shocks, and dashed lines represent funded shocks. Blue lines indicate transfers to the bottom 10 percent of the liquid-asset distribution, and red lines indicate transfers to the top 10 percent. In each case, each recipient receives the same amount.

investment. Transfers to the bottom 10 percent generate a much larger rise in consumption than transfers to the top 10 percent. Alternatively, unfunded transfers to the bottom 10 percent and to the top 10 percent generate very similar responses in those aggregate variables.

When transfers are funded, the main channel is the direct spending response of the recipients. Households at the bottom of the liquid-asset distribution have higher marginal propensities to consume, whereas households at the top have lower marginal propensities to consume. Transfers to the bottom 10 percent therefore raise consumption much more strongly, whereas transfers to the top 10 percent produce a much smaller increase in consumption and a weaker aggregate response.

When transfers are unfunded, fiscal backing adds a second force to this MPC heterogene-

ity. Households become less willing to hold nominal government bonds at the pre-shock price level, demand shifts toward goods, and inflation rises. That portfolio-reallocation channel pushes up output and inflation even when the transfer is targeted to the top 10 percent. As a result, targeting matters less for the aggregate responses to unfunded transfers than it does when transfers are funded. At the same time, the consumption response remains stronger when the transfer goes to the bottom 10 percent, because the direct spending channel is still larger for households with low liquid assets.

This comparison shows that targeting and fiscal backing affect transmission in different ways. Targeting mainly changes the direct spending response of the recipients. Fiscal backing mainly changes whether the transfer also induces a broader shift away from nominal government bonds and into goods. Once the transfer is unfunded, that second channel becomes strong enough that transfers to low- and high-liquid-asset households generate similar movements in output and inflation.

3.4 Role of Portfolio Adjustment Opportunities

This section studies how the funded–unfunded difference depends on households’ ability to rebalance between liquid nominal assets and illiquid capital. The key parameter is the portfolio adjustment probability λ , which governs the fraction of households that can adjust illiquid asset holdings in a given period. A higher value of λ means that households have more frequent access to the illiquid capital market and can move wealth between liquid nominal assets and illiquid real assets more easily.

Figure 4 compares impulse responses for two values of λ : the baseline value, $\lambda = 0.03$, and a lower-adjustment case, $\lambda = 0.005$. The comparison shows that portfolio adjustment opportunities matter primarily for the difference between funded and unfunded deficits. When households can adjust illiquid assets more frequently, the quantitative gap between funded and unfunded responses becomes larger. Unfunded deficits generate stronger responses, while funded deficits generate weaker responses. This pattern appears for both

government-purchases shocks and transfer shocks.

The reason is that changing λ affects two margins at once. First, more frequent portfolio adjustment improves households' ability to self-insure against idiosyncratic income risk. When households can access illiquid capital more easily, they rely less on current disposable income to smooth consumption. This dampens the effects of funded deficits, because funded deficits are backed by higher expected future primary surpluses and do not generate a large revaluation of nominal government debt. In this case, easier access to illiquid assets mainly strengthens households' ability to smooth consumption over time, so the responses of consumption, output, investment, and inflation become smaller.

Second, more frequent portfolio adjustment strengthens the response to unfunded deficits. When fiscal backing is weak, households want to reduce their exposure to nominal government liabilities. A higher λ makes this reallocation easier: a larger fraction of households can move out of liquid nominal assets and into illiquid real capital in response to the shock. This stronger rebalancing raises investment directly and amplifies the decline in the real value of nominal government debt. As a result, unfunded deficits generate larger responses of investment, output, consumption, and inflation.

A higher λ also changes steady-state portfolio composition. When households can adjust illiquid assets more frequently, they need to hold fewer liquid nominal assets for precautionary purposes and can hold more wealth in illiquid capital. The steady-state liquid-to-illiquid asset ratio therefore falls. This portfolio composition matters for the transmission of fiscal shocks. In the higher-adjustment economy, households enter the shock with a smaller reliance on liquid nominal assets for self-insurance, but they also have a stronger ability to rebalance toward real capital when fiscal backing weakens. These two effects work in opposite directions for funded and unfunded shocks: they dampen funded responses but amplify unfunded responses.

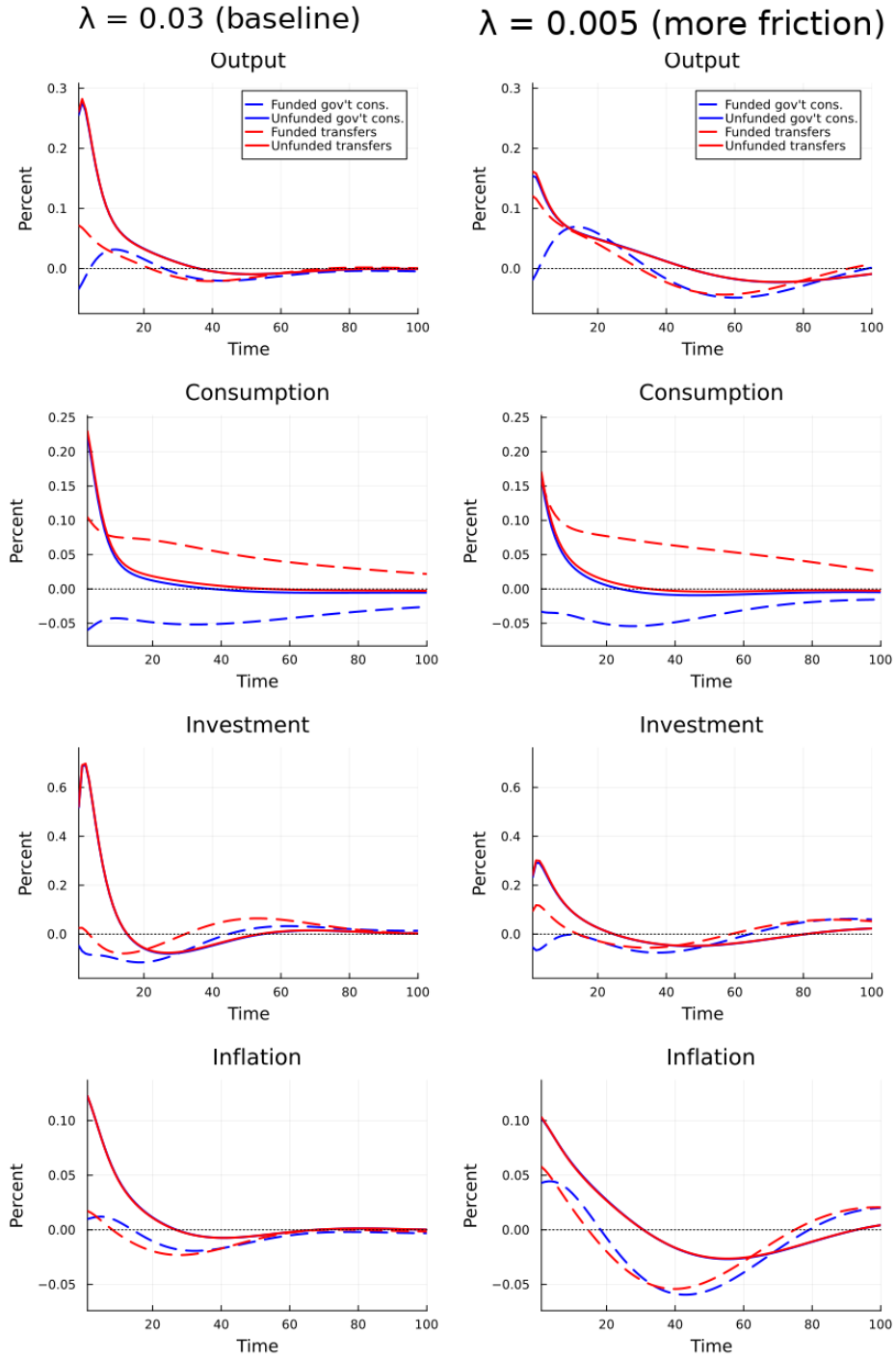


Figure 4: Impulse responses under different portfolio adjustment probabilities. The left column reports responses under $\lambda = 0.03$, and the right column reports responses under $\lambda = 0.005$. Rows show output, consumption, investment, and inflation. A higher λ enlarges the difference between funded and unfunded responses.

These results support the paper’s main mechanism. The inflationary effects of fiscal deficits depend not only on household heterogeneity, but also on the assets households can use to respond to fiscal news. When households can reallocate wealth across liquid nominal bonds and illiquid real capital, weak fiscal backing triggers a portfolio response that is absent or muted in environments with only liquid nominal assets. The fiscal theory mechanism therefore remains useful for understanding inflation in heterogeneous-agent economies, because portfolio choice determines how strongly households can move away from nominal government liabilities when deficits are not backed.

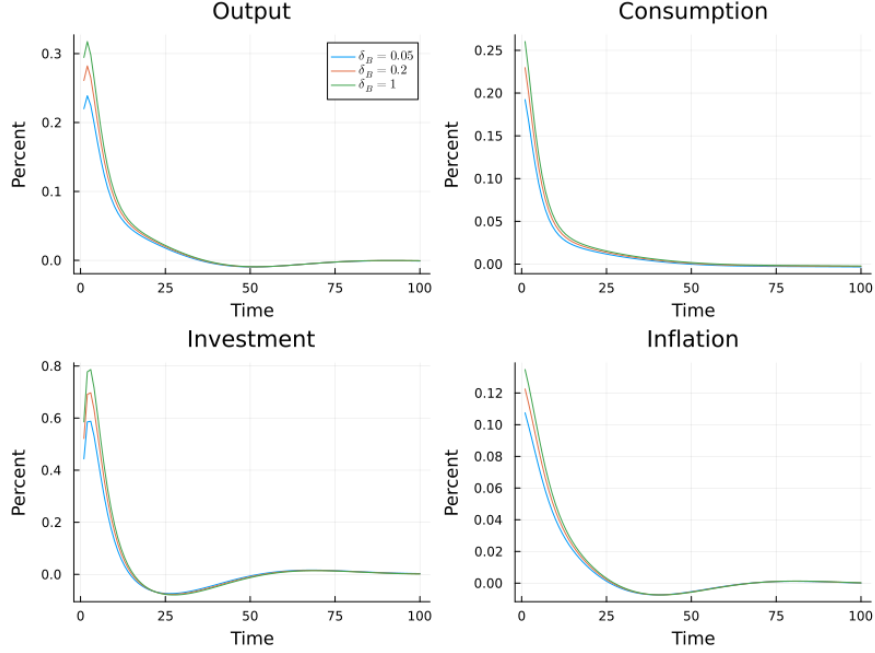
3.5 Role of Debt Maturity

The maturity structure of government debt matters because it affects the valuation of outstanding nominal liabilities. Since the paper’s mechanism operates through changes in the real value of nominal government debt and households’ portfolio reallocation away from that debt, debt maturity could in principle change the strength of the response to unfunded fiscal shocks. Longer-maturity debt makes a larger share of government liabilities sensitive to the expected path of inflation and nominal interest rates, while short-term debt is repriced more quickly.

Figure 5 reports the responses to an unfunded transfer shock for the baseline value $\delta_B = 0.2$ and two alternative values, $\delta_B = 0.05$ and $\delta_B = 1$, where δ_B is the fraction of long-term debt that matures each period. Lower values of δ_B correspond to longer maturity. When $\delta_B = 1$, all debt matures each period, so the government issues only one-period debt. Debt maturity has a visible but quantitatively limited effect on the responses to unfunded transfers. A longer debt maturity structure produces smaller responses of output, consumption, investment, and inflation. These differences do not change the paper’s main quantitative results: unfunded fiscal shocks remain substantially more expansionary and inflationary than funded shocks, and the assumed maturity structure of government debt is

not the main driver of the results.[†]

Figure 5: Role of debt maturity: unfunded transfers



Note: Impulse Response Functions (IRFs) for output, consumption, investment, and inflation following an unfunded transfer shock, for alternative values of δ_B , the fraction of long-term debt maturing each period.

4 Conclusion

In this paper, I study under what circumstances fiscal deficits become inflationary in a heterogeneous-agent New Keynesian economy with long-term nominal debt and endogenous household portfolio choice. The key distinction is whether deficits are backed by higher expected future primary surpluses. When backing is weak, households reduce their demand for nominal government bonds and rebalance toward goods and real assets. This portfolio response raises inflation and real activity by reducing the real value of outstanding nominal liabilities and shifting demand away from government debt.

[†]For funded transfer shocks and funded government-purchases shocks, the responses are less sensitive to alternative values of δ_B ; the differences across debt-maturity specifications are very small.

The central result is that household portfolio choice changes how fiscal backing is transmitted. In an environment where households hold both liquid nominal government liabilities and illiquid real capital, weak fiscal backing does not only lower the real value of nominal debt. It also induces households to rebalance away from nominal liabilities and toward real assets. This reallocation amplifies the effects of unfunded deficits on consumption, investment, output, and inflation.

This mechanism helps explain why unfunded deficits generate much larger aggregate responses than funded deficits. It also explains why unfunded government purchases and unfunded transfers generate very similar aggregate responses. Once fiscal backing is weak, the common portfolio-reallocation channel becomes quantitatively important. When deficits are funded, by contrast, expected future primary surpluses limit the revaluation of nominal debt, portfolio reallocation is muted, and the fiscal instrument matters more. Funded transfers stimulate private spending more than funded government purchases because they provide direct support to households with higher marginal propensities to consume.

Fiscal backing also shapes distributional outcomes. Unfunded deficits reduce the real value of nominal asset positions and lower consumption inequality by raising consumption among households with low and middle liquid wealth while reducing consumption among households with large nominal positions. Funded deficits do not generate the same revaluation effect, so their distributional consequences depend more on the fiscal instrument and the allocation of transfers.

The portfolio margin is central to these results. When households can adjust illiquid asset positions more easily, the gap between funded and unfunded deficits becomes larger. Easier access to illiquid capital allows households to smooth consumption more effectively when deficits are funded, but it also strengthens the reallocation away from nominal government bonds when fiscal backing weakens. Thus, the fiscal theory mechanism remains useful for understanding inflation because portfolio choice determines how strongly households can move away from nominal government liabilities when deficits are not backed.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the author used ChatGPT, an AI-assisted writing tool developed by OpenAI, to assist with language editing and improving clarity of exposition. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the manuscript.

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A Additional Model Details

This appendix describes the preference, income, tax, production, and aggregation blocks that support the model in the main text.

A.1 Preferences, Labor Supply, and Tax Progressivity

Households have Greenwood–Hercowitz–Huffman (GHH) preferences. Household i chooses consumption c_{it} and labor supply n_{it} to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{it} - G(h_{it}, n_{it})), \quad (28)$$

where β is the discount factor, h_{it} is idiosyncratic labor productivity, and $G(h_{it}, n_{it})$ is the disutility of labor. Period utility is

$$u(x_{it}) = \frac{x_{it}^{1-\xi} - 1}{1-\xi}, \quad (29)$$

with

$$x_{it} = c_{it} - G(h_{it}, n_{it}), \quad (30)$$

where ξ is the risk-aversion parameter.

Following [Benabou \(2002\)](#) and [Heathcote et al. \(2017\)](#), after-tax labor income is

$$y_{it} = (1 - \tau_t^L) (w_t h_{it} n_{it})^{1-\tau_t^P}, \quad (31)$$

where w_t is the aggregate real wage, τ_t^L is the labor-tax level, and τ_t^P is tax progressivity. A higher value of τ_t^P implies a more progressive tax schedule.

The marginal disutility of labor is

$$\frac{\partial G(h_{it}, n_{it})}{\partial n_{it}} = (1 - \tau_t^P)(1 - \tau_t^L)(w_t h_{it})^{1-\tau_t^P} n_{it}^{-\tau_t^P} = (1 - \tau_t^P) \frac{y_{it}}{n_{it}}. \quad (32)$$

Assume that the disutility function has constant elasticity with respect to labor supply:

$$\frac{\partial G(h_{it}, n_{it})}{\partial n_{it}} = (1 + \gamma) \frac{G(h_{it}, n_{it})}{n_{it}}, \quad (33)$$

with $\gamma > 0$. Then the composite consumption-labor term can be written as

$$x_{it} = c_{it} - \frac{1 - \tau_t^P}{1 + \gamma} y_{it}. \quad (34)$$

Thus, household decisions can be expressed in terms of the transformed object x_{it} and after-tax income y_{it} .

For the quantitative solution, I follow [Bayer et al. \(2024\)](#) and keep tax progressivity fixed at its steady-state value in the labor-supply block, $\tau_t^P \equiv \bar{\tau}^P$. I use

$$G(h_{it}, n_{it}) = h_{it}^{1-\bar{\tau}^P} \frac{n_{it}^{1+\gamma}}{1 + \gamma}. \quad (35)$$

Under this specification, the labor-supply first-order condition implies that individual productivity does not affect optimal hours. All employed households choose the same labor supply, denoted by N_t .

Given the normalization $\int h_{it} di \equiv 1$, total effective labor input is

$$\int h_{it} n_{it} di = N_t. \quad (36)$$

Substituting the common labor choice into after-tax labor income gives

$$y_{it} = (1 - \tau_t^L)(w_t h_{it} n_{it})^{1-\bar{\tau}^P} \quad (37)$$

$$= (1 - \tau_t^L)^{\frac{1+\gamma}{\gamma+\bar{\tau}^P}} (1 - \bar{\tau}^P)^{\frac{1-\bar{\tau}^P}{\gamma+\bar{\tau}^P}} w_t^{\frac{1+\gamma}{\gamma+\bar{\tau}^P} (1-\bar{\tau}^P)} h_{it}^{1-\bar{\tau}^P}. \quad (38)$$

Income heterogeneity therefore comes from heterogeneity in labor productivity.

A.2 Household Income, Occupation, and Profit Distribution

The household income process follows [Bayer et al. \(2024\)](#). I distinguish between workers and entrepreneurs. Workers face idiosyncratic labor-income risk. Entrepreneurs correspond to

the zero-labor state and receive non-union firm profits.

Let $\tilde{h}_{it}$ denote pre-normalized productivity. Productivity evolves as

$$\tilde{h}_{it} = \begin{cases} \exp(\rho_h \log \tilde{h}_{it-1} + \epsilon_{it}^h) & \text{with probability } 1 - \zeta \text{ if } h_{it-1} \neq 0, \\ 1 & \text{with probability } \iota \text{ if } h_{it-1} = 0, \\ 0 & \text{otherwise,} \end{cases} \quad (39)$$

where

$$h_{it} = \frac{\tilde{h}_{it}}{\int \tilde{h}_{it} di}. \quad (40)$$

Households with $h_{it} \neq 0$ are workers, while households with $h_{it} = 0$ are entrepreneurs. Worker productivity follows a log-AR(1) process with persistence ρ_h and innovation $\epsilon_{it}^h \sim \mathcal{N}(0, \bar{\sigma}_h^2)$. The parameter ζ governs transitions from worker to entrepreneur, and ι governs transitions from entrepreneur to worker.

Workers receive labor income and a lump-sum share of union profits. Entrepreneurs receive firm profits other than union profits. They can liquidate only a fraction ω^Π of firm profits through tradable profit claims with stochastic maturity. A fraction ι^Π of existing claims matures each period. Let q_t^Π denote the price of profit-claim paper. Entrepreneurial profit income is

$$\Pi_t^E = (1 - \omega^\Pi)\Pi_t^F + \iota^\Pi q_t^\Pi, \quad (41)$$

where Π_t^F denotes firm profits.

Using

$$c_{it} = x_{it} + \frac{1 - \bar{\tau}^P}{1 + \gamma} y_{it}, \quad (42)$$

the household budget constraint can be written as

$$\begin{aligned} x_{it} + b_{it+1} + q_t k_{it+1} &= \frac{R_{it}}{\pi_t} b_{it} + (q_t + r_t) k_{it} + \frac{\bar{\tau}^P + \gamma}{1 + \gamma} y_{it} \\ &+ \mathbb{I}_{h_{it} \neq 0} (1 - \tau_t) \Pi_t^U + \mathbb{I}_{h_{it} = 0} (1 - \tau_t^L) (\Pi_t^E)^{1 - \bar{\tau}^P} + Transfers_{it}, \end{aligned} \quad (43)$$

subject to

$$k_{it+1} \geq 0, \quad b_{it+1} \geq \underline{B}. \quad (44)$$

A.3 Tax Revenue Construction

The average tax rate τ_t in the main text is the ratio of tax revenue to total pre-tax labor and entrepreneurial income. Let w_t denote the real wage, n_{it} hours worked, and h_{it} idiosyncratic productivity. The average tax rate is

$$\tau_t = \frac{\mathbb{E}_t[w_t n_{it} h_{it} + \mathbb{I}_{h_{it}=0} \Pi_t^E] - \tau_t^L \mathbb{E}_t[(w_t h_{it} n_{it} + \mathbb{I}_{h_{it}=0} \Pi_t^E)^{1-\tau_t^P}]}{\mathbb{E}_t[w_t n_{it} h_{it} + \mathbb{I}_{h_{it}=0} \Pi_t^E]}. \quad (45)$$

The level parameter τ_t^L adjusts to match the average tax rate implied by the tax rule in the main text.

Aggregate tax revenue is

$$Tax_t = \tau_t \mathbb{E}_t[w_t n_{it} h_{it} + \mathbb{I}_{h_{it} \neq 0} \Pi_t^U + \mathbb{I}_{h_{it}=0} \Pi_t^E], \quad (46)$$

where Π_t^U denotes union profits and Π_t^E denotes entrepreneurial profits.

A.4 Firm-Side Problem

A.4.1 Final Goods Producers

Final goods producers aggregate differentiated intermediate goods y_{jt} into final output Y_t using a CES aggregator:

$$Y_t = \left(\int y_{jt}^{\frac{\eta_t-1}{\eta_t}} dj \right)^{\frac{\eta_t}{\eta_t-1}}, \quad (47)$$

where η_t is the elasticity of substitution across varieties. The associated price index is

$$P_t = \left(\int p_{jt}^{1-\eta_t} dj \right)^{\frac{1}{1-\eta_t}}, \quad (48)$$

and demand for variety j is

$$y_{jt} = \left(\frac{p_{jt}}{P_t} \right)^{-\eta_t} Y_t. \quad (49)$$

A.4.2 Intermediate Goods Producers

Intermediate goods producers operate under monopolistic competition and produce with Cobb–Douglas technology:

$$y_{jt} = Z_t N_{jt}^\alpha (u_{jt} K_{jt})^{1-\alpha}, \quad (50)$$

where Z_t is total factor productivity, N_{jt} is labor, K_{jt} is capital, and u_{jt} is utilization.

Capital depreciation increases with utilization:

$$\delta(u_{jt}) = \delta_0 + \delta_1(u_{jt} - 1) + \frac{\delta_2}{2}(u_{jt} - 1)^2. \quad (51)$$

Firm j minimizes

$$w_t^F N_{jt} + (r_t + q_t \delta(u_{jt})) K_{jt},$$

which yields

$$w_t^F = \alpha mc_{jt} Z_t \left(\frac{u_{jt} K_{jt}}{N_{jt}} \right)^{1-\alpha}, \quad (52)$$

$$r_t + q_t \delta(u_{jt}) = u_{jt} (1 - \alpha) mc_{jt} Z_t \left(\frac{N_{jt}}{u_{jt} K_{jt}} \right)^\alpha, \quad (53)$$

and

$$q_t [\delta_1 + \delta_2(u_{jt} - 1)] = (1 - \alpha) mc_{jt} Z_t \left(\frac{N_{jt}}{u_{jt} K_{jt}} \right)^\alpha. \quad (54)$$

Under symmetry, marginal costs are equalized across firms: $mc_{jt} = mc_t$.

Price setting is subject to Calvo frictions. Each period, a firm can reset its price with probability $1 - \lambda_Y$. A reoptimizing firm maximizes the expected discounted value of profits:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \lambda_Y^t (1 - \tau_t^L) Y_t^{1-\tau_t^P} \left\{ \left(\frac{p_{jt} \bar{\pi}^t}{P_t} - mc_t \right) \left(\frac{p_{jt} \bar{\pi}^t}{P_t} \right)^{-\eta_t} \right\}^{1-\tau_t^P}. \quad (55)$$

The price Phillips curve is

$$\log\left(\frac{\pi_t}{\bar{\pi}}\right) = \beta \mathbb{E}_t \log\left(\frac{\pi_{t+1}}{\bar{\pi}}\right) + \kappa_Y \left(mc_t - \frac{1}{\mu_t^Y}\right), \quad (56)$$

where

$$\kappa_Y = \frac{(1 - \lambda_Y)(1 - \beta\lambda_Y)}{\lambda_Y}, \quad \frac{1}{\mu_t^Y} = \frac{\eta_t - 1}{\eta_t}. \quad (57)$$

A.4.3 Capital Goods Producers

Capital goods producers take the price of capital q_t as given and choose investment I_t to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t I_t \left\{ \Psi_t q_t \left[1 - \frac{\phi}{2} \left(\log \frac{I_t}{I_{t-1}} \right)^2 \right] - 1 \right\}, \quad (58)$$

where Ψ_t is the marginal efficiency of investment and ϕ governs investment adjustment costs.

The optimality condition for investment is

$$\Psi_t q_t \left[1 - \phi \log \left(\frac{I_t}{I_{t-1}} \right) \right] = 1 - \beta \mathbb{E}_t \left[\Psi_{t+1} q_{t+1} \phi \log \left(\frac{I_{t+1}}{I_t} \right) \right]. \quad (59)$$

Capital evolves according to

$$K_t = (1 - \delta(u_t)) K_{t-1} + \Psi_t \left[1 - \frac{\phi}{2} \left(\log \frac{I_t}{I_{t-1}} \right)^2 \right] I_t. \quad (60)$$

A.4.4 Labor Packers and Unions

Households sell labor to unions, which supply differentiated labor services to labor packers.

Labor packers aggregate these services into total labor input:

$$N_t = \left(\int \hat{n}_{jt}^{\frac{\zeta_t-1}{\zeta_t}} dj \right)^{\frac{\zeta_t}{\zeta_t-1}}, \quad (61)$$

where $\zeta_t > 1$ is the elasticity of substitution across labor types. Demand for labor type j is

$$\hat{n}_{jt} = \left(\frac{W_{jt}}{W_t^F} \right)^{-\zeta_t} N_t, \quad (62)$$

where W_{jt} is the nominal wage set by union j and W_t^F is the nominal wage paid by labor packers.

Unions face Calvo wage rigidity. Each period, a union can reset its wage with probability $1 - \lambda_w$. A reoptimizing union maximizes

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \lambda_w \frac{W_t^F}{P_t} N_t \left\{ \left(\frac{W_{jt} \bar{\pi}_W^t}{W_t^F} - \frac{W_t}{W_t^F} \right) \left(\frac{W_{jt} \bar{\pi}_W^t}{W_t^F} \right)^{-\zeta_t} \right\}. \quad (63)$$

The wage Phillips curve is

$$\log \left(\frac{\pi_t^W}{\bar{\pi}_W} \right) = \beta \mathbb{E}_t \log \left(\frac{\pi_{t+1}^W}{\bar{\pi}_W} \right) + \kappa_w \left(mc_t^w - \frac{1}{\mu_t^W} \right), \quad (64)$$

where

$$\pi_t^W = \frac{W_t^F}{W_{t-1}^F} = \frac{w_t^F}{w_{t-1}^F} \pi_t, \quad mc_t^w = \frac{w_t}{w_t^F}, \quad \frac{1}{\mu_t^W} = \frac{\xi_t - 1}{\xi_t}, \quad (65)$$

and

$$\kappa_w = \frac{(1 - \lambda_w)(1 - \beta \lambda_w)}{\lambda_w}. \quad (66)$$

B Additional Calibration Details

This appendix reports the full steady-state calibration, implementation constants, policy-rule parameters, shock processes, and additional remarks on the parameters most closely tied to the mechanism.

B.1 Steady-State Calibration

I pin down the steady state by fixing a set of preference, technology, income-process, liquidity/portfolio, and fiscal parameters so that the model matches the same broad quarterly targets as [Bayer et al. \(2024\)](#). The targeted objects include earnings persistence and dispersion, worker–entrepreneur flows, the liquid–illiquid composition of household wealth, borrower prevalence, factor shares, government size, and tax progressivity.

Table 3: Steady-state calibration: parameters and targets (quarterly)

Block	Symbol	Value	Target / Mapping	Data / Source
<i>Preferences and labor supply</i>				
Risk aversion	ξ	4.000	Precautionary saving strength	Kaplan–Violante (2014)
Inverse Frisch elasticity	γ	2.000	Labor supply slope	Chetty et al. (2011)
Discount factor	β	0.98255	K/Y	NIPA long-run averages
Labor disutility scale	γ_{scale}	0.200	Steady hours in $[0.3, 0.4]$	BLS/NIPA (hours)
<i>Income process and occupation</i>				
Persistence (labor inc.)	ρ_h	0.980	Earnings persistence	Storesletten–Telmer–Yaron (2004)
Std. dev. (labor inc.)	σ_h	0.120	Earnings dispersion	Storesletten–Telmer–Yaron (2004)
$E \rightarrow W$ transition	ι	$1/16 = 0.0625$	Worker share in steady state	Güvenen–Kaplan–Song (2014)
$W \rightarrow E$ transition	ζ	$1/4500 \approx 2.22 \times 10^{-4}$	Upper-tail wealth concentration	WID/SCF

Continued on next page

Table 3: Steady-state calibration: parameters and targets (quarterly)
(continued)

Block	Symbol	Value	Target / Mapping	Data / Source
<i>Technology and markups</i>				
Capital share	α	0.318	Labor share ≈ 0.68	NIPA factor shares
Depreciation (quarterly)	δ_0	0.0215	K/Y	NIPA CFC
Price markup (gross)	μ^Y	1.10	Micro price-cost margins	Standard benchmark
Wage markup (gross)	μ^W	1.10	Micro wage markup	Standard benchmark
<i>Liquidity, maturity, and asset supply</i>				
Portfolio adj. prob.	λ	0.030	Liquid/illiquid turnover	SCF/NIPA (as in BBL)
Bond maturity	δ_B	0.200	Avg. duration = $1/\delta_B$	Treasury duration stylized fact
Bonds-to-capital	ψ	0.100	$(B + \bar{q}_\Pi)/K$	SCF/NIPA aggregate ratio
Tradable profit share	ω_Π	0.150	Liquid equity-like paper	BBL steady-state mapping
Retirement of profit shares	ι_Π	0.016	Liquid claim turnover	BBL steady-state mapping
<i>Fiscal block and normalizations</i>				
$1 - \tau^L$	$1 - \tau^L$	0.645	$G/Y \approx 0.20$ in steady state	NIPA government share
Tax progressivity	τ^P	0.1022	Average progressivity	SOI
Gross inflation	π	1.000	Normalization (indexation)	Convention
Gross policy rate	R^b	1.000	Normalization (quarterly)	Convention

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Table 4: Implementation constants and scenario controls

Symbol	Value	Role	
ASHIFT	1.000	Borrowing-wedge shift	Implementation scale
$\omega_{\Pi}^{\text{bar}}$	0.20	Profit-claim scaling	Complements $(\omega_{\Pi}, \iota_{\Pi})$ mapping
TRANSFERtoG	1.00	Baseline Transfer/ G ratio	Scenario toggle
BFRatio, GFRatio	0.95, 0.95	Tax / adjustment shares	Fiscal decomposition
α_G	0.0	Government spending in utility	Scenario toggle

Table 3: Steady-state calibration: parameters and targets (quarterly)
(continued)

Block	Symbol	Value	Target / Mapping	Data / Source
Borrower wedge	$\bar{R}$	0.0466	Borrower share $\sim 16\%$	SCF
Return on capital	R	1.010	Scale consistency with β, δ_0	Normalization

Notes: The steady-state calibration targets the same broad long-run empirical objects as [Bayer et al. \(2024\)](#): income persistence and dispersion, worker–entrepreneur flows, the combined liquid paper supply $(B + \bar{q}_{\Pi})$ relative to illiquid capital K , the prevalence of borrowers, factor shares, government size, and tax progressivity. “BBL steady-state mapping” refers to the accounting identity linking $(\omega_{\Pi}, \iota_{\Pi})$ to $\bar{q}_{\Pi}/Y$ in their steady state. Inflation and the gross policy rate are normalized under indexation. The borrower wedge is $\bar{R} = \pi(1.20)^{1/4} - 1$.

B.2 Implementation Constants and Scenario Controls

A small number of parameters act as scenario controls or implementation constants. These objects do not discipline the steady-state targets in Table 3, but they enter the quantitative implementation.

B.3 Policy Rules and Shock Processes

This section reports the policy-rule parameters and exogenous shock processes used in the benchmark simulations. The funded and unfunded fiscal rules are the main policy objects for the paper’s mechanism. The remaining shocks complete the quantitative environment.

Table 5: Policy-rule and shock-process parameters (calibrated baseline)

Block	Symbol	Value	Symbol	Value	Symbol	Value
<i>Exogenous aggregate shocks</i>						
Spread / wedge	ρ_A	0.90	σ_A	0.001		
TFP	ρ_Z	0.90	σ_Z	0.001		
Investment-good productivity	ρ_Ψ	0.90	σ_Ψ	0.001		
Price markup	ρ_{μ^Y}	0.90	σ_{μ^Y}	0.001		
Wage markup	ρ_{μ^W}	0.90	σ_{μ^W}	0.001		
<i>Income-risk process</i>						
Income-risk state	ρ_s	0.84	σ_s	0.001		
Cyclical sensitivity	Σ_N	0.00				
<i>Monetary policy</i>						
Taylor rule	ρ_R	0.70	θ_π	0.50	θ_Y	0.125
Monetary shock	σ_R	0.001				
<i>Fiscal policy: tax rules</i>						
Avg. tax rate	ρ_τ	0.50	γ_B^τ	0.00	γ_Y^τ	0.00
Progressivity	ρ_P	0.50	γ_B^P	0.00	γ_Y^P	0.00
Progressivity shock	σ_P	0.001				
<i>Fiscal components: purchases and transfers</i>						

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Table 5: Policy-rule and shock-process parameters (calibrated baseline) (*continued*)

Block	Symbol	Value	Symbol	Value	Symbol	Value
GF shock	$\rho_{GFshock}$	0.98	$\sigma_{GFshock}$	0.001		
GU shock	$\rho_{GUshock}$	0.98	$\sigma_{GUshock}$	0.001		
TF shock	$\rho_{TFshock}$	0.98	$\sigma_{TFshock}$	0.001		
TU shock	$\rho_{TUshock}$	0.98	$\sigma_{TUshock}$	0.001		
GF rule	ρ_{GF}	0.90	γ_{YGF}	0.50	γ_{BGF}	-0.50
GU rule	ρ_{GU}	0.90	γ_{YGU}	0.50	γ_{BGU}	0.50
TF rule	ρ_{TF}	0.90	γ_{YTF}	0.50	γ_{BTF}	-0.50
TU rule	ρ_{TU}	0.90	γ_{YTU}	0.50	γ_{BTU}	0.50

Notes: Values report the calibrated baseline used in the simulations. Government purchases and transfers are decomposed as $G_t = GU_t + GF_t$ and $Transfer_t = TU_t + TF_t$. Total long-term nominal government debt is decomposed as $Bgov_t = BFgov_t + BUGov_t$, where $BUGov_t \equiv Bgov_t - BFgov_t$. The rules for GF_t and TF_t feed back on $BFgov_t$, while the rules for GU_t and TU_t feed back on $BUGov_t$. Negative coefficients on funded debt imply stabilizing future adjustment, while nonnegative coefficients on unfunded debt imply the absence of debt-stabilizing backing.

C Numerical Solution Methods

I solve the model using a Reiter-style algorithm. The procedure has four steps. I first compute the stationary equilibrium by solving the household problems and the invariant distribution jointly. I then reduce the dimensionality of the cross-sectional distribution. Next, I linearize the equilibrium conditions around the stationary equilibrium. Finally, I solve the resulting linear rational-expectations system by generalized Schur decomposition and compute impulse responses.

C.1 Overview of the Computational Approach

Let the individual state be (m, k, s) , where m denotes the liquid position, k denotes illiquid real capital, and s collects idiosyncratic income and occupation states. Aggregate states include nominal rigidities, fiscal and monetary policy variables, taxes, and the cross-sectional distribution $\Theta_t(m, k, s)$.

The computation proceeds as follows:

1. Construct grids for (m, k) and a Markov chain for s .
2. Solve the household block for adjusters and non-adjusters. Adjusters choose both m' and k' . Non-adjusters choose m' and keep k fixed.
3. Iterate the distribution forward using the household policy rules and the idiosyncratic transition matrix.
4. Reduce the distributional state, linearize the equilibrium system, and solve the linearized model using QZ/Schur methods.

C.2 Discretization: Idiosyncratic Risk and Asset Grids

Idiosyncratic states. Worker productivity follows a log-AR(1) process. I approximate it by a finite-state Markov chain with n_s states using a Tauchen-style method. Occupation switching between workers and entrepreneurs is incorporated by adding an entrepreneur state and modifying the transition matrix. Let $\Phi \in \mathbb{R}^{n_s \times n_s}$ denote the resulting transition matrix.

Liquid and illiquid grids. Liquid assets m are discretized on a non-uniform grid with more points near the borrowing constraint $\underline{m} < 0$. Illiquid capital k is discretized on a non-uniform grid with more points at low levels of capital and fewer points in the upper tail. Denote the grid sizes by n_m and n_k .

Portfolio adjustment. Each period, a fraction λ of households can adjust illiquid capital. With probability $1 - \lambda$, a household keeps $k' = k$ and chooses only m' . The household block therefore yields two liquid-saving policies, $m'_a(m, k, s)$ and $m'_n(m, k, s)$, and one capital policy, $k'(m, k, s)$, for adjusters.

C.3 Household Problem and EGM Implementation

C.3.1 Recursive Formulation

Let $W_t(m, k, s)$ denote the continuation value before the capital-adjustment draw. The conditional value functions are

$$V_t^a(m, k, s) = \max_{m', k'} u(x_t) + \beta \mathbb{E} [W_{t+1}(m', k', s') \mid s], \quad (67)$$

$$V_t^n(m, k, s) = \max_{m'} u(x_t) + \beta \mathbb{E} [W_{t+1}(m', k, s') \mid s], \quad (68)$$

$$W_t(m, k, s) = \lambda V_t^a(m, k, s) + (1 - \lambda) V_t^n(m, k, s), \quad (69)$$

subject to the budget constraint and the constraints on (m', k') . The choice variable entering utility is the GHH composite x_t .

C.3.2 Euler Equations and the EGM Step

The liquid Euler equation is

$$u_x(x_t) = \beta \mathbb{E}_t [u_x(x_{t+1}) R_{t+1}^m], \quad (70)$$

where R_{t+1}^m is the real gross return on the liquid position, including the borrowing wedge when $m < 0$.

For a candidate next-period state (m', k', s') , define the expected marginal value of liquid wealth as

$$\mathcal{M}_t(m', k', s) := \mathbb{E} [u_x(x_{t+1}) R_{t+1}^m \mid s]. \quad (71)$$

Then (70) implies

$$u_x(x_t) = \beta \mathcal{M}_t(m', k', s) \quad \Rightarrow \quad x_t = u_x^{-1}(\beta \mathcal{M}_t(m', k', s)). \quad (72)$$

Non-adjusters. For non-adjusters, $k' = k$. Combining (72) with the budget constraint gives the endogenous current liquid position associated with each m' :

$$m = \frac{m' + q_t k' + x_t - \text{income}_t(m, k, s)}{R_t^m}, \quad (73)$$

where $\text{income}_t(\cdot)$ collects labor income, profit income, transfers, and returns on k . I impose $m' \geq \underline{m}$ and use the standard EGM correction at the borrowing constraint.

Adjusters. For adjusters, the household chooses both m' and k' . I use a two-asset EGM that combines a grid search over k' with an EGM step over m' :

1. Fix a candidate k' on the illiquid grid.
2. Given k' , apply the liquid EGM step and construct the implied x_t and endogenous m grid.
3. Compute the implied value for each candidate k' and take the upper envelope:

$$V_t^a(m, k, s) = \max_{k' \in \mathcal{K}} \tilde{V}_t^a(m, k, s; k').$$

The upper-envelope step handles the non-concavities that arise from discrete adjustment of illiquid capital and the borrowing constraint.

Interpolation. The EGM step produces policies on an endogenous grid. I map policies back to the exogenous (m, k) grid using monotone piecewise-linear interpolation.

C.4 Forward Iteration for the Cross-Sectional Distribution

Given policy functions and the idiosyncratic transition matrix Φ , the distribution evolves through a discrete transition operator. Let Θ_t denote the probability mass on the tensor grid $\mathcal{M} \times \mathcal{K} \times \mathcal{S}$. For each current grid point (m, k, s) , the policy-implied next states are

$$(m'_a, k') \quad \text{for adjusters,} \quad (m'_n, k) \quad \text{for non-adjusters.}$$

If a policy-implied state lies between grid points, I split mass across adjacent points using linear interpolation weights. I then apply the idiosyncratic transition matrix Φ and combine adjusters and non-adjusters:

$$\Theta_{t+1} = \lambda \mathcal{T}_a(\Theta_t) + (1 - \lambda) \mathcal{T}_n(\Theta_t),$$

where $\mathcal{T}_a$ and $\mathcal{T}_n$ are the transition operators implied by the two sets of policy functions.

Stationary distribution. The invariant distribution $\bar{\Theta}$ solves $\bar{\Theta} = \mathcal{T}(\bar{\Theta})$. I compute it by forward iteration with damping until the distribution is numerically stable and total mass is preserved.

C.5 Stationary Equilibrium

A stationary equilibrium consists of prices, policy variables, household policies, and an invariant distribution such that households optimize, firms and banks optimize, government budget identities and fiscal rules hold, and all markets clear.

Outer loop. I solve for the steady-state price and policy vector using a damped fixed-point or Broyden routine:

1. Guess the steady-state price and policy vector.

2. Solve the household block.
3. Compute the invariant distribution.
4. Aggregate household decisions and evaluate market-clearing residuals.
5. Update the price and policy vector.

Long-term nominal bond pricing. Long-term government debt has geometric maturity δ_B . In steady state, the no-arbitrage condition implies

$$\bar{q}^B = \frac{1}{\bar{R}^b - (1 - \delta_B)}. \quad (74)$$

I use this identity in the steady-state computation and as a check on the bond-pricing block.

C.6 Distribution Reduction for Linear Dynamics

The full tensor distribution $\Theta_t \in \mathbb{R}^{n_m n_k n_s}$ is too large to include directly in the linearized system. I therefore approximate the distribution using a discrete cosine transform (DCT).

DCT representation. Let θ denote a vectorized object on a grid of length n , and let $D \in \mathbb{R}^{n \times n}$ be the DCT matrix. The transformed coefficients are

$$\hat{\theta} = D\theta.$$

Low-frequency coefficients capture smooth movements in the distribution, while high-frequency coefficients capture local curvature near kinks and constraints.

Coefficient selection. I retain the smallest set of coefficients that captures a fixed share of the squared ℓ_2 norm:

$$\frac{\|\hat{\theta}_I\|_2}{\|\hat{\theta}\|_2} \geq 1 - \text{reduc_value}.$$

The retained index set $\mathcal{I}$ is constructed by ordering coefficients by magnitude.

Reduced state vector. Let S denote the selection matrix that maps the full DCT coefficients into the retained subset:

$$\tilde{\theta} = S\hat{\theta}.$$

The linearized system uses $\tilde{\theta}$ and the aggregate state variables as the state vector. When reconstructing probability objects, I apply the inverse DCT and renormalize the distribution to preserve total mass.

C.7 Linearization and Solution of the Rational-Expectations System

Stack the aggregate equilibrium conditions and the reduced distribution law into

$$\mathbb{E}_t [F(Z_{t+1}, Z_t, \varepsilon_{t+1})] = 0, \tag{75}$$

where Z_t contains aggregate states, aggregate controls, and reduced distribution coefficients.

Linearizing around the steady state gives

$$A \Delta Z_{t+1} = B \Delta Z_t + C \varepsilon_{t+1}. \tag{76}$$

I solve (76) using generalized Schur decomposition. The solution yields policy functions of the form

$$\Delta Z_{t+1} = H \Delta Z_t + M \varepsilon_{t+1}, \quad \Delta X_t = G \Delta Z_t,$$

where ΔX_t denotes selected controls.

C.8 Impulse Responses

Given (H, M) , impulse responses are computed by iterating

$$\Delta Z_{t+1} = H\Delta Z_t, \quad t \geq 1, \quad \Delta Z_1 = M\varepsilon_1.$$

All reported impulse responses are scaled to a common fiscal innovation size.

C.9 Diagnostics and Numerical Accuracy

The implementation records the following diagnostics:

1. **Euler residuals.** Liquid Euler equation errors in consumption units, separately for borrowers and savers.
2. **Market clearing.** Residuals in the liquid asset market, capital market, and government budget identities.
3. **Mass conservation.** Total mass of the cross-sectional distribution and non-negativity of probability weights.
4. **Grid sensitivity.** Stability of the main aggregate responses under refinements of (n_m, n_k, n_s) and the DCT reduction threshold.

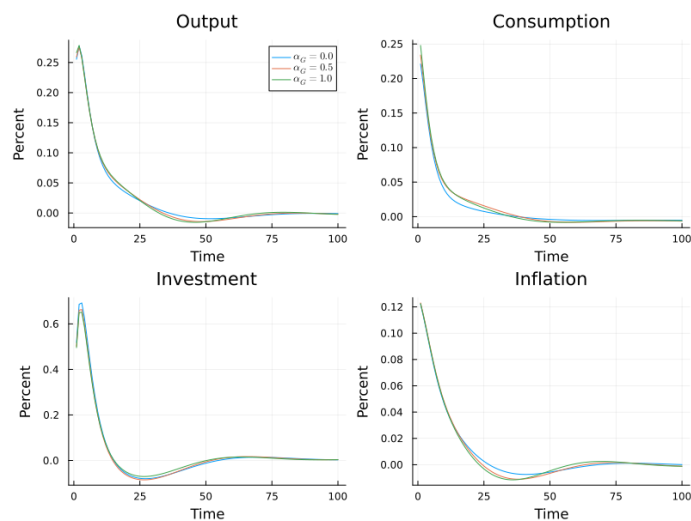
Implementation note. The solution is written in `Julia`. The code uses separate routines for Markov-chain construction, EGM policy updates for adjusters and non-adjusters, forward iteration of the distribution, DCT-based reduction and reconstruction, and Schur/QZ solution of the linearized system.

D Robustness Checks

D.1 Robustness to Government Spending in Utility

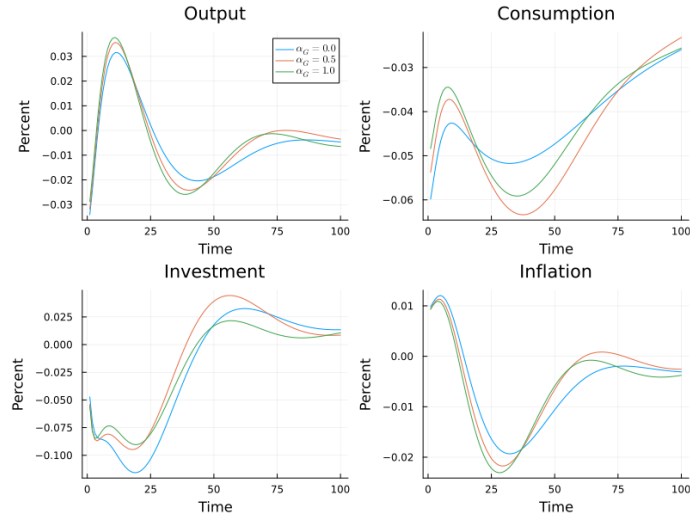
This appendix allows government spending to enter household utility directly through $c + \alpha_G G$. I consider three values of α_G : 0, 0.5, and 1.0. Across all four fiscal shocks, the responses of output, consumption, investment, and inflation remain quantitatively similar. The main results are therefore robust to this modification of preferences.

Figure 6: Robustness to government spending in utility: unfunded government purchases



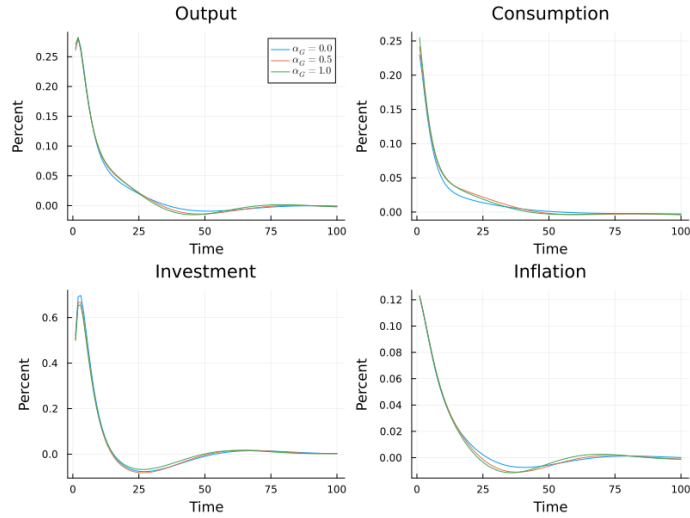
Note: Impulse Response Functions (IRFs) for output, consumption, investment, and inflation following an unfunded government-purchases shock, for alternative values of α_G .

Figure 7: Robustness to government spending in utility: funded government purchases



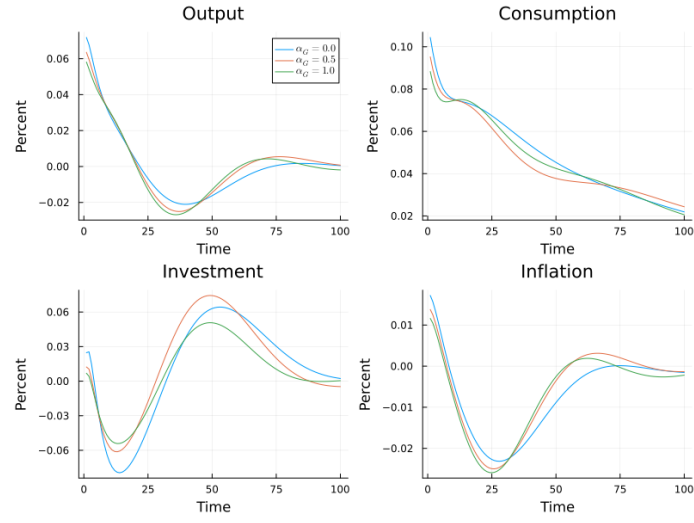
Note: Impulse Response Functions (IRFs) for output, consumption, investment, and inflation following a funded government-purchases shock, for alternative values of α_G .

Figure 8: Robustness to government spending in utility: unfunded transfers



Note: Impulse Response Functions (IRFs) for output, consumption, investment, and inflation following an unfunded transfer shock, for alternative values of α_G .

Figure 9: Robustness to government spending in utility: funded transfers



Note: Impulse Response Functions (IRFs) for output, consumption, investment, and inflation following a funded transfer shock, for alternative values of α_G .